

Yifan Li

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Introduction

I am a first year PhD student advised by Prof. Yoshihiro KAWAHARA at The University of Tokyo. I'm interested in wearable devices, human-computer interaction.

Education

PhD	The University of Tokyo , Electrical Engineering and Information Systems • Advisor: Prof. Yoshihiro Kawahara	Sept 2025 – Present
MS	The University of Tokyo , Electrical Engineering and Information Systems • Thesis: Ultra-low-powered wireless input ring and machine-knitable plug-and-play e-textiles	Oct 2023 – Sept 2025
BS	University of Glasgow , Electronics and Electrical Engineering, UESTC joint program • Honours of the First Class • Thesis: Research and Implementation of Federated Learning Algorithms for Non-shared(Non-IID) Data Scenarios	Sept 2019 – Sept 2023

Experience

Apple , Systems EE Intern in AirPods • TBD	Shanghai, China Jan 2026 – Present
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Projects

picoRing mouse <i>Principal researcher</i>	Oct. 2024 - Apr. 2025
- Developed an ultra-low-powered (μW-level) tiny mouse ring that supports subtle thumb-to-index scrolling and pressing interactions in real-world wearable computing situations and XR interactions. - Tools Used: C, Python, PyQt, data fitting, electrical RF circuit design and measurement. - The result has been published to UIST 2025.	
Plug-and-Play e-knit <i>Principal researcher</i>	Mar. 2024 - Oct. 2024
- Developed plug-n-play e-knit, a large-scale, reconfigurable, scalable e-textile prototyping tool, supporting magnetically-repositionable e-textile prototyping. - Tools Used: C, Python, PyQt, signal processing, electrical circuit design and prototyping. - The result has been published to TEI 2025.	

Publications

- 1) **Y. Li**, M. Fukumoto, M. Kari, S. Ishida, A. Noda, T. Yokota, T. Someya, Y. Kawahara, R. Takahashi, Ultra-low-power ring-based wireless tinymouse. *Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology (UIST '25)*, Article 17, 1–12. <https://doi.org/10.1145/3746059.3747615>
- 2) **Y. Li**, R. Takahashi, W. Yukita, K. Matsutani, C. Caremel, Y. Iwamoto, S. Lee, T. Yokota, T. Someya, Y. Kawahara. Plug-n-play e-knit: prototyping large-area e-textiles using machine-knitted magnetically-repositionable sensor networks. *In Proceedings of the Nineteenth International Conference on Tangible, Embedded, and Embodied Interaction (TEI '25)*. Article 56, 1–7, Mar. 2025. <https://doi.org/10.1145/3689050.3705973>

- 3) **Y. Li**, M. Fukumoto, M. Kari, T. Yokota, T. Someya, Y. Kawahara, R. Takahashi. Demo of picoRing mouse: ultra-low-powered wireless mouse ring with ring-to-wristband coil-based impedance sensing. *In Proceedings of the Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*. Article 707, 1–5, Apr. 2025. <https://doi.org/10.1145/3706599.3721183>

Awards

The University of Tokyo Fellowship	Oct 2023
Outstanding student award with grant-in-aid (~33000 USD) from The University of Tokyo	
Best Master's Thesis	Sept 2025
Outstanding master thesis award from The University of Tokyo	
World-leading Innovative Graduate Study Program “Co-designing Future Society”	Apr 2024
UTokyo's high-level research education program with financial support (~30000 USD) and research grant (~10000 USD)	
Best Bachelor's Degree Thesis.	Sept 2023
Outstanding bachelor thesis award from The University of Glasgow	

Skills

Key words: Wearable computing, Sensing, Ubiquitous computing, UIST

Language

- Chinese: Native
- English: Fluent, IELTS: 7.5
- Japanese: Intermediate

Hardware

- Electrical circuit design: Ultra-low-power circuits, RF circuits for wireless power transfer and inductive sensing, sensing circuits, etc., designed for mass production.
- RF measurements: Vector network analyzer, impedance analyzer, impedance matching, frequency response analyzer, etc.
- Embedded Programming: Programming embedded systems (ARM, HAL).

Software

- CAD tools: Autodesk EAGLE, Altium Designer, Kicad, Autodesk Fusion, etc.
- Programming languages: C, Python, MATLAB, etc.
- Computer vision: Experience in implementing Visual SLAM using OpenCV.
- Simulation: Experience in MATLAB, LTSpice, Ansys HFSS, etc.
- Others: Experience in: Adobe CC (Illustrator, Photoshop, Premiere), MS Office (Excel, PowerPoint, Word).